

Module Handbook – M.Sc. Industrial Artificial Intelligence

Stand: 15. September 2025

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Semester 1

This section contains 6 module descriptions.

IAI-M-01 Mathematical Foundations of Machine Learning

Semester	1	ECTS	6	SWS	4
Workload	180 h (60 h contact, 120 h self-study)	Frequency	Annually, winter semester	Duration	1 semester
Module type	Compulsory	Language	English	Module coordinator	Prof. Dr. Helene Trautwein

Formal prerequisites: None

Recommended prior knowledge: Undergraduate linear algebra and calculus

Learning outcomes

On successful completion of the module, students are able to

- reformulate applied problems in the language of linear algebra and multivariate calculus
- derive gradients of composite functions and apply them to parameter estimation
- assess the numerical stability of matrix decompositions used in learning algorithms
- evaluate probabilistic models using the language of measure-theoretic probability at an applied level

Module content

- Vector spaces, norms, inner products; orthogonal projection
- Eigenvalue and singular value decomposition; low-rank approximation
- Multivariate differentiation, Jacobians, Hessians; chain rule for computational graphs
- Convexity, first- and second-order optimality conditions
- Gradient descent, stochastic gradient descent, momentum methods; convergence behaviour
- Probability spaces, random vectors, conditional expectation, maximum likelihood estimation
- Matrix calculus for backpropagation

Teaching and learning methods: Lecture (2 SWS) with integrated exercise sessions (2 SWS); weekly problem sets discussed in class

Form of assessment: K120 — Written examination, 120 minutes

Requirements for the award of credit points: Passing the written examination (K120). Credit points are awarded once the module examination has been passed.

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Deisenroth, Faisal, Ong: Mathematics for Machine Learning, Cambridge University Press
- Boyd, Vandenberghe: Convex Optimization, Cambridge University Press
- Strang: Linear Algebra and Learning from Data, Wellesley-Cambridge Press

Usability: Compulsory module of the M.Sc. Industrial Artificial Intelligence. Provides the mathematical basis for IAI-M-03 and IAI-M-07.

Module description IAI-M-01 — coordinator: Prof. Dr. Helene Trautwein, Faculty of Engineering and Computer Science.

IAI-M-02 Industrial Data Engineering

Semester	1	ECTS	6	SWS	4
Workload	180 h (60 h contact, 120 h self-study)	Frequency	Annually, winter semester	Duration	1 semester
Module type	Compulsory	Language	English	Module coordinator	Prof. Dr.-Ing. Kaspar Rehnitz

Formal prerequisites: None

Recommended prior knowledge: Programming experience in Python; basic relational database knowledge

Learning outcomes

On successful completion of the module, students are able to

- design ingestion pipelines for high-frequency machine and process data
- select appropriate storage models for time-series, event and relational industrial data
- implement data quality checks and quantify their effect on downstream model performance
- document data lineage in a form suitable for audit

Module content

- Industrial data sources: PLC, SCADA, MES, ERP; OPC UA information models
- Time-series storage, downsampling, retention policies
- Batch and streaming ingestion; message brokers
- Schema evolution and data contracts
- Data quality dimensions; drift detection at the ingestion layer
- Data lineage, provenance and reproducibility
- Practical work with a simulated production line data set

Teaching and learning methods: Lecture (2 SWS), laboratory exercise (2 SWS) in small groups on the faculty's data engineering testbed

Form of assessment: PJ — Project work; documented data pipeline with written report

- Implemented pipeline and documentation (70 %)
- Oral defence of design decisions (approx. 15 min) (30 %)

Requirements for the award of credit points: Submission and acceptance of the project work including the oral defence. Both components must be passed independently.

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system; laboratory equipment and experimental set-ups

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Kleppmann: Designing Data-Intensive Applications, O'Reilly
- Reis, Housley: Fundamentals of Data Engineering, O'Reilly
- OPC Foundation: OPC Unified Architecture specification parts 1–5

Usability: Compulsory module of the M.Sc. Industrial Artificial Intelligence. Prerequisite for IAI-M-09.

Module description IAI-M-02 — coordinator: Prof. Dr.-Ing. Kaspar Rehnitz, Faculty of Engineering and Computer Science.

IAI-M-03 Machine Learning I: Supervised and Unsupervised Methods

Semester	1	ECTS	6	SWS	4
Workload	180 h (60 h contact, 120 h self-study)	Frequency	Annually, winter semester	Duration	1 semester
Module type	Compulsory	Language	English	Module coordinator	Prof. Dr. Helene Trautwein

Formal prerequisites: None

Recommended prior knowledge: Parallel attendance of IAI-M-01

Learning outcomes

On successful completion of the module, students are able to

- select learning algorithms appropriate to a given industrial problem and data regime
- design validation strategies that avoid optimistic bias under temporal correlation
- interpret model behaviour using feature attribution methods
- quantify and communicate model uncertainty to non-specialist stakeholders

Module content

- Empirical risk minimisation; bias-variance decomposition
- Linear and logistic regression; regularisation
- Support vector machines and kernel methods
- Decision trees, random forests, gradient boosting
- Clustering, density estimation, dimensionality reduction
- Cross-validation under temporal and group structure
- Model interpretability: permutation importance, SHAP, partial dependence

Teaching and learning methods: Lecture (3 SWS) with computer-based exercises (1 SWS)

Form of assessment: K120 — Written examination, 120 minutes

Requirements for the award of credit points: Passing the written examination (K120).

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system; computer workstations and specialist software

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Hastie, Tibshirani, Friedman: The Elements of Statistical Learning, Springer
- Murphy: Probabilistic Machine Learning – An Introduction, MIT Press
- Molnar: Interpretable Machine Learning

Usability: Compulsory module of the M.Sc. Industrial Artificial Intelligence. Prerequisite for IAI-M-07 and IAI-M-08.

Module description IAI-M-03 — coordinator: Prof. Dr. Helene Trautwein, Faculty of Engineering and Computer Science.

IAI-M-04 Sensor Systems and Signal Processing

Semester	1	ECTS	6	SWS	4
Workload	180 h (60 h contact, 120 h self-study)	Frequency	Annually, winter semester	Duration	1 semester
Module type	Compulsory	Language	English	Module coordinator	Prof. Dr.-Ing. Anselm Bierbaum

Formal prerequisites: None

Recommended prior knowledge: Fundamentals of electrical engineering; Fourier analysis

Learning outcomes

On successful completion of the module, students are able to

- characterise industrial sensors by their transfer behaviour, noise and failure modes
- design digital filters for the suppression of process-typical disturbances
- apply time-frequency methods to non-stationary machine signals
- justify sampling and quantisation decisions against a measurement requirement

Module content

- Sensor principles: piezoelectric, inductive, optical, thermal
- Metrological characteristics: sensitivity, hysteresis, drift, calibration
- Sampling, aliasing, quantisation noise
- FIR and IIR filter design
- Spectral estimation; windowing; short-time Fourier transform
- Wavelet analysis for transient detection
- Condition indicators for rotating machinery

Teaching and learning methods: Lecture (3 SWS) and laboratory exercise (1 SWS) on the vibration analysis test rig

Form of assessment: K90 — Written examination, 90 minutes

Requirements for the award of credit points: Passing the written examination (K90). Successful completion of the laboratory exercise is a prerequisite for admission to the examination.

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system; laboratory equipment and experimental set-ups

Admission requirement: Successful completion of the laboratory exercise is a prerequisite for admission to the examination.

Literature:

- Oppenheim, Schaffer: Discrete-Time Signal Processing, Pearson
- Randall: Vibration-based Condition Monitoring, Wiley
- Tränkler, Reindl: Sensortechnik, Springer Vieweg

Usability: Compulsory module of the M.Sc. Industrial Artificial Intelligence. Prerequisite for IAI-M-13.

Module description IAI-M-04 — coordinator: Prof. Dr.-Ing. Anselm Bierbaum, Faculty of Engineering and Computer Science.

IAI-M-05 Research Methods and Scientific Writing

Semester	1	ECTS	3	SWS	2
Workload	90 h (30 h contact, 60 h self-study)	Frequency	Annually, winter semester	Duration	1 semester
Module type	Compulsory	Language	English	Module coordinator	Prof. Dr. Marit Sonnleitner

Formal prerequisites: None

Recommended prior knowledge: None

Learning outcomes

On successful completion of the module, students are able to

- formulate a research question that is answerable within the constraints of a master's thesis
- conduct a structured literature search and appraise source quality
- design an experimental protocol including controls and statistical power considerations
- write in the register and structure expected of an engineering research report

Module content

- Research designs in applied engineering; internal and external validity
- Systematic literature search; bibliographic management
- Statistical hypothesis testing, effect sizes, multiple comparison
- Reproducibility, pre-registration, data availability
- Academic integrity, citation practice, plagiarism
- Structure and register of technical reports

Teaching and learning methods: Seminar (2 SWS) with peer review of drafts

Form of assessment: RE — Presentation with written elaboration

Requirements for the award of credit points: Presentation of approximately 20 minutes and written elaboration of 8 to 12 pages. Both parts form a single examination performance and are assessed together.

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Booth, Colomb, Williams: The Craft of Research, University of Chicago Press
- Wohlin et al.: Experimentation in Software Engineering, Springer

Usability: Compulsory module of the M.Sc. Industrial Artificial Intelligence. Prepares the master's thesis (IAI-M-21).

Module description IAI-M-05 — coordinator: Prof. Dr. Marit Sonnleitner, Faculty of Engineering and Computer Science.

IAI-M-06 German as a Foreign Language A1

Semester	1	ECTS	3	SWS	4
Workload	90 h (60 h contact, 30 h self-study)	Frequency	Annually, winter semester	Duration	1 semester
Module type	Compulsory	Language	German/English	Module coordinator	Language Centre, Dr. Ilka Wendehorst

Formal prerequisites: None

Recommended prior knowledge: No prior knowledge of German required

Learning outcomes

On successful completion of the module, students are able to

- understand and use familiar everyday expressions and very basic phrases
- introduce themselves and others and ask and answer questions about personal details
- interact in a simple way in the university and workplace context
- read short workplace safety notices with dictionary support

Module content

- Greetings, introductions, numbers, time
- Present tense of regular and common irregular verbs
- Nominative and accusative case; definite and indefinite articles
- Everyday and university vocabulary; basic workplace vocabulary
- Simple written correspondence

Teaching and learning methods: Language course (4 SWS) in groups of no more than 18 participants

Form of assessment: PF — Portfolio comprising at least three components

- Written test (40 %)
- Oral interaction task (30 %)
- Learning diary (30 %)

Requirements for the award of credit points: Portfolio comprising the three components listed above; each component must be submitted, and the aggregate must be passed.

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Course materials provided by the Language Centre

Usability: Compulsory module of the M.Sc. Industrial Artificial Intelligence. Students who provide evidence of German language competence at level A2 or above of the Common European Framework may apply to the examination board for recognition under § 15 ASPO.

Note: Unusual workload structure: 4 SWS contact time against 3 ECTS, reflecting the high contact-hour requirement of beginner language instruction.

Module description IAI-M-06 — coordinator: Language Centre, Dr. Ilka Wendehorst, Language Centre.

Semester 2

This section contains 5 module descriptions.

IAI-M-07 Deep Learning for Industrial Applications

Semester	2	ECTS	6	SWS	4
Workload	180 h (60 h contact, 120 h self-study)	Frequency	Annually, summer semester	Duration	1 semester
Module type	Compulsory	Language	English	Module coordinator	Prof. Dr. Helene Trautwein

Formal prerequisites: Successful completion of IAI-M-03

Recommended prior knowledge: IAI-M-01

Learning outcomes

On successful completion of the module, students are able to

- design network architectures matched to the structure of industrial data
- diagnose and remedy training pathologies
- apply transfer learning where labelled industrial data are scarce
- estimate the inference cost of a trained model against deployment constraints

Module content

- Feedforward networks, backpropagation, initialisation
- Regularisation: dropout, weight decay, early stopping, data augmentation
- Convolutional architectures; residual connections
- Sequence models: recurrent networks, temporal convolutions, attention
- Transfer learning and fine-tuning under limited labelled data
- Model compression: pruning, quantisation, distillation
- Training diagnostics and reproducibility

Teaching and learning methods: Lecture (2 SWS) and computer laboratory (2 SWS) on the faculty GPU cluster

Form of assessment: K120 — Written examination, 120 minutes

Requirements for the award of credit points: Passing the written examination (K120).

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system; laboratory equipment and experimental set-ups; computer workstations and specialist software

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Goodfellow, Bengio, Courville: Deep Learning, MIT Press
- Zhang et al.: Dive into Deep Learning, Cambridge University Press

Usability: Compulsory module of the M.Sc. Industrial Artificial Intelligence. Prerequisite for IAI-M-15 and IAI-M-18.

Module description IAI-M-07 — coordinator: Prof. Dr. Helene Trautwein, Faculty of Engineering and Computer Science.

IAI-M-08 Industrial Computer Vision

Semester	2	ECTS	6	SWS	4
Workload	180 h (60 h contact, 120 h self-study)	Frequency	Annually, summer semester	Duration	1 semester
Module type	Compulsory	Language	English	Module coordinator	Prof. Dr.-Ing. Anselm Bierbaum

Formal prerequisites: Successful completion of IAI-M-03

Recommended prior knowledge: None

Learning outcomes

On successful completion of the module, students are able to

- specify an industrial vision system including illumination and optics
- implement classical and learning-based inspection pipelines
- evaluate inspection performance against defect rates and cost of error
- assess the limits of vision-based inspection in a given production context

Module content

- Image formation, optics, illumination geometries
- Camera calibration and photometric correction
- Classical feature detection, segmentation, morphology
- Learning-based detection and segmentation
- Anomaly detection with few defect samples
- Performance evaluation under class imbalance; cost-sensitive thresholds
- Integration with production line control

Teaching and learning methods: Lecture (2 SWS) and project laboratory (2 SWS) at the vision inspection cell

Form of assessment: PJ — Project work

Requirements for the award of credit points: Submission and acceptance of a working inspection pipeline with written report of 15 to 20 pages.

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system; laboratory equipment and experimental set-ups

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Szeliski: Computer Vision – Algorithms and Applications, Springer
- Steger, Ulrich, Wiedemann: Machine Vision Algorithms and Applications, Wiley-VCH

Usability: Compulsory module of the M.Sc. Industrial Artificial Intelligence.

Module description IAI-M-08 — coordinator: Prof. Dr.-Ing. Anselm Bierbaum, Faculty of Engineering and Computer Science.

IAI-M-09 Cyber-Physical Production Systems

Semester	2	ECTS	6	SWS	4
Workload	180 h (60 h contact, 120 h self-study)	Frequency	Annually, summer semester	Duration	1 semester
Module type	Compulsory	Language	English	Module coordinator	Prof. Dr.-Ing. Kaspar Rehnitz

Formal prerequisites: Successful completion of IAI-M-02

Recommended prior knowledge: None

Learning outcomes

On successful completion of the module, students are able to

- describe the automation hierarchy and its dissolution in modern architectures
- specify communication requirements for latency-critical control loops
- assess the safety and security implications of networked production equipment
- design the interface between an analytics component and a control system

Module content

- Automation pyramid; edge, fog and cloud allocation
- Fieldbus and industrial Ethernet; time-sensitive networking
- Real-time constraints; determinism and jitter
- Asset administration shells and digital representation
- Functional safety in networked systems; IEC 61508 overview
- Industrial security: segmentation, IEC 62443 overview
- Interfacing analytics with control: advisory versus closed-loop operation

Teaching and learning methods: Lecture (3 SWS) and laboratory (1 SWS) on the model production line

Form of assessment: K90 — Written examination, 90 minutes

Requirements for the award of credit points: Passing the written examination (K90).

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system; laboratory equipment and experimental set-ups

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Vogel-Heuser, Bauernhansl, ten Hompel (eds.): Handbuch Industrie 4.0, Springer Vieweg
- Sauter et al.: Industrial Communication Technology Handbook, CRC Press

Usability: Compulsory module of the M.Sc. Industrial Artificial Intelligence.

Module description IAI-M-09 — coordinator: Prof. Dr.-Ing. Kaspar Rehnitz, Faculty of Engineering and Computer Science.

IAI-M-10 Master Project Industrial AI

Semester	2	ECTS	6	SWS	2
Workload	180 h (30 h contact, 150 h self-study)	Frequency	Annually, summer semester	Duration	1 semester
Module type	Compulsory	Language	English	Module coordinator	Programme director, Prof. Dr. Helene Trautwein

Formal prerequisites: Successful completion of at least 24 ECTS credits from the first semester of the programme

Recommended prior knowledge: IAI-M-02, IAI-M-03

Learning outcomes

On successful completion of the module, students are able to

- plan and execute an applied AI project against an externally supplied requirement
- allocate work within a team and manage technical dependencies
- produce a solution that is deployable in the client's technical environment
- present results to a mixed technical and managerial audience

Module content

- Project definition in cooperation with an industrial partner or research group
- Requirements elicitation and scoping
- Iterative development with milestone reviews
- Technical documentation and handover
- Final presentation to the client

Teaching and learning methods: Project work (2 SWS supervision) with three scheduled milestone reviews

Form of assessment: PJ — Project work, carried out in teams of three to five students

Requirements for the award of credit points: Acceptance of the team project report and the final presentation. Individual contributions must be separately identifiable and are assessed individually.

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Project-specific; determined at project definition

Usability: Compulsory module of the M.Sc. Industrial Artificial Intelligence.

Note: Deviating from § 20 Abs. 1 ASPO, this module may be repeated only once (§ 9 Abs. 1 BSPO IAI).

Module description IAI-M-10 — coordinator: Programme director, Prof. Dr. Helene Trautwein, Faculty of Engineering and Computer Science.

IAI-M-11 Ethics, Law and Safety of AI Systems

Semester	2	ECTS	3	SWS	2
Workload	90 h (30 h contact, 60 h self-study)	Frequency	Annually, summer semester	Duration	1 semester
Module type	Compulsory	Language	English	Module coordinator	Prof. Dr. Marit Sonnleitner

Formal prerequisites: None

Recommended prior knowledge: None

Learning outcomes

On successful completion of the module, students are able to

- identify the regulatory regime applicable to an AI system in an industrial setting
- conduct a structured risk assessment for a deployed model
- articulate the ethical trade-offs in automation decisions affecting workers
- document conformity considerations for a high-risk application

Module content

- European regulatory framework for artificial intelligence; risk classes
- Product safety and machinery regulation; interaction with AI components
- Data protection in industrial contexts; works council participation
- Fairness, accountability and transparency in employment-adjacent systems
- Liability allocation for autonomous decisions
- Case studies from manufacturing and logistics

Teaching and learning methods: Seminar (2 SWS) with case discussion

Form of assessment: RE — Presentation with written elaboration

Requirements for the award of credit points: Presentation of approximately 20 minutes and written elaboration of 8 to 12 pages.

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Wachter, Mittelstadt, Floridi: Transparent, Explainable and Accountable AI
- Current consolidated texts of the applicable European regulations

Usability: Compulsory module of the M.Sc. Industrial Artificial Intelligence.

Module description IAI-M-11 — coordinator: Prof. Dr. Marit Sonnleitner, Faculty of Engineering and Computer Science.

Semester 3

This section contains 1 module description.

IAI-M-21 Master's Thesis

Semester	3	ECTS	30	SWS	0
Workload	900 h (0 h contact, 900 h self-study)	Frequency	Every semester	Duration	1 semester
Module type	Compulsory	Language	English	Module coordinator	Chair of the examination board

Formal prerequisites: Successful completion of at least 54 ECTS credits of the programme

Recommended prior knowledge: IAI-M-05

Learning outcomes

On successful completion of the module, students are able to

- work independently on a research or development problem to a professional standard
- situate the work within the relevant state of the art
- defend methodological choices against alternatives
- report results in a form suitable for publication or industrial handover

Module content

- Independent treatment of a topic from the field of industrial artificial intelligence
- Topic agreed with a supervising examiner and registered with the examination board

Teaching and learning methods: Independent work under supervision; supervision meetings at least monthly

Form of assessment: MA — Master's thesis; processing time six months

- Written thesis (100 %)

Requirements for the award of credit points: Submission of the thesis within the processing period and assessment as at least sufficient (4.0).

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Topic-specific

Usability: Compulsory module of the M.Sc. Industrial Artificial Intelligence. Concludes the programme.

Note: § 13 Abs. 4 BSPO IAI provides for a colloquium of approximately 30 minutes as part of this module. § 33 Abs. 3 Satz 2 ASPO, inserted by the Second Amendment Statute, provides that no colloquium takes place in master's programmes of fewer than 120 ECTS credits. See documented conflict K-04.

Module description IAI-M-21 — coordinator: Chair of the examination board, Faculty of Engineering and Computer Science.

Elective catalogue

This section contains 9 module descriptions.

IAI-M-12 Reinforcement Learning for Process Control

Semester	2	ECTS	3	SWS	2
Workload	90 h (30 h contact, 60 h self-study)	Frequency	Annually, summer semester	Duration	1 semester
Module type	Elective	Language	English	Module coordinator	Prof. Dr. Helene Trautwein

Formal prerequisites: None

Recommended prior knowledge: IAI-M-03

Learning outcomes

On successful completion of the module, students are able to

- formulate a control problem as a Markov decision process
- select between value-based and policy-based methods for a given problem
- assess the sample efficiency and safety implications of learning in a physical plant

Module content

- Markov decision processes; dynamic programming
- Temporal difference learning; Q-learning
- Policy gradient methods; actor-critic
- Offline reinforcement learning from logged process data
- Safe exploration and constrained policies

Teaching and learning methods: Lecture with integrated exercises (2 SWS)

Form of assessment: M20 — Oral examination, approx. 20 minutes

Requirements for the award of credit points: Passing the oral examination (M20).

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Sutton, Barto: Reinforcement Learning – An Introduction, MIT Press

Usability: Elective module of the M.Sc. Industrial Artificial Intelligence, elective catalogue under Annex 2 of the Special Regulations.

Module description IAI-M-12 — coordinator: Prof. Dr. Helene Trautwein, Faculty of Engineering and Computer Science.

IAI-M-13 Time Series Analysis and Predictive Maintenance

Semester	2	ECTS	3	SWS	2
Workload	90 h (30 h contact, 60 h self-study)	Frequency	Annually, summer semester	Duration	1 semester
Module type	Elective	Language	English	Module coordinator	Prof. Dr.-Ing. Anselm Bierbaum

Formal prerequisites: Successful completion of IAI-M-04

Recommended prior knowledge: IAI-M-03

Learning outcomes

On successful completion of the module, students are able to

- model degradation processes from condition monitoring data
- estimate remaining useful life with calibrated uncertainty
- design a maintenance policy that balances intervention and failure cost

Module content

- Stationarity, autocorrelation, spectral representation
- ARIMA and state space models
- Survival analysis and degradation modelling
- Remaining useful life estimation; prognostic metrics
- Maintenance policy optimisation

Teaching and learning methods: Lecture (1 SWS) and computer laboratory (1 SWS)

Form of assessment: PJ — Project work

Requirements for the award of credit points: Submission and acceptance of the project work.

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system; laboratory equipment and experimental set-ups; computer workstations and specialist software

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Hyndman, Athanasopoulos: Forecasting – Principles and Practice
- Si et al.: Remaining Useful Life Estimation – A Review

Usability: Elective module of the M.Sc. Industrial Artificial Intelligence.

Module description IAI-M-13 — coordinator: Prof. Dr.-Ing. Anselm Bierbaum, Faculty of Engineering and Computer Science.

IAI-M-14 Natural Language Processing in Engineering Contexts

Semester	2	ECTS	3	SWS	2
Workload	90 h (30 h contact, 60 h self-study)	Frequency	Annually, summer semester	Duration	1 semester
Module type	Elective	Language	English	Module coordinator	Prof. Dr. Marit Sonnleitner

Formal prerequisites: None

Recommended prior knowledge: IAI-M-03

Learning outcomes

On successful completion of the module, students are able to

- process unstructured maintenance and quality records
- build retrieval systems over technical documentation
- assess the reliability of language models in safety-relevant documentation tasks

Module content

- Text preprocessing for technical German and English
- Representation: embeddings, contextual models
- Information extraction from maintenance logs
- Retrieval over technical documentation
- Evaluation and failure modes in specialist domains

Teaching and learning methods: Lecture (1 SWS) and computer laboratory (1 SWS)

Form of assessment: PJ — Project work

Requirements for the award of credit points: Submission and acceptance of the project work.

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system; laboratory equipment and experimental set-ups; computer workstations and specialist software

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Jurafsky, Martin: Speech and Language Processing

Usability: Elective module of the M.Sc. Industrial Artificial Intelligence.

Module description IAI-M-14 — coordinator: Prof. Dr. Marit Sonnleitner, Faculty of Engineering and Computer Science.

IAI-M-15 Edge AI and Embedded Inference

Semester	2	ECTS	3	SWS	2
Workload	90 h (30 h contact, 60 h self-study)	Frequency	Annually, summer semester	Duration	1 semester
Module type	Elective	Language	English	Module coordinator	Prof. Dr.-Ing. Kaspar Rehnitz

Formal prerequisites: Successful completion of IAI-M-07

Recommended prior knowledge: None

Learning outcomes

On successful completion of the module, students are able to

- deploy a trained model to a resource-constrained embedded target
- measure and optimise inference latency and energy consumption
- select quantisation strategies appropriate to an accuracy budget

Module content

- Embedded accelerator architectures
- Quantisation-aware training and post-training quantisation
- Runtime environments for embedded inference
- Latency and energy measurement methodology
- Over-the-air model update strategies

Teaching and learning methods: Laboratory course (2 SWS) on embedded evaluation boards

Form of assessment: LB — Laboratory work with certificate; ungraded

Requirements for the award of credit points: Successful completion of all five laboratory assignments. The module is assessed as passed or not passed.

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system; laboratory equipment and experimental set-ups

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Warden, Situnayake: TinyML, O'Reilly

Usability: Elective module of the M.Sc. Industrial Artificial Intelligence.

***Note:** This module is assessed as an ungraded Studienleistung and does not contribute to the overall grade (§ 11 Abs. 3 BSPO IAI).*

Module description IAI-M-15 — coordinator: Prof. Dr.-Ing. Kaspar Rehnitz, Faculty of Engineering and Computer Science.

IAI-M-16 Digital Twins and Simulation

Semester	2	ECTS	3	SWS	2
Workload	90 h (30 h contact, 60 h self-study)	Frequency	Annually, summer semester	Duration	1 semester
Module type	Elective	Language	English	Module coordinator	Prof. Dr.-Ing. Kaspar Rehnitz

Formal prerequisites: None

Recommended prior knowledge: IAI-M-09

Learning outcomes

On successful completion of the module, students are able to

- distinguish digital model, digital shadow and digital twin
- couple a simulation model to live process data
- validate a simulation against measured plant behaviour

Module content

- Simulation paradigms: discrete event, continuous, hybrid
- Model calibration and validation
- Co-simulation and interface standards
- Synchronisation of twin and physical asset
- Use of twins for what-if analysis and training data generation

Teaching and learning methods: Lecture (1 SWS) with simulation exercises (1 SWS)

Form of assessment: RE — Presentation with written elaboration

Requirements for the award of credit points: Presentation of approximately 20 minutes and written elaboration of 8 to 12 pages.

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Law: Simulation Modeling and Analysis, McGraw-Hill

Usability: Elective module of the M.Sc. Industrial Artificial Intelligence.

Module description IAI-M-16 — coordinator: Prof. Dr.-Ing. Kaspar Rehnitz, Faculty of Engineering and Computer Science.

IAI-M-17 MLOps and Model Lifecycle Management

Semester	2	ECTS	3	SWS	2
Workload	90 h (30 h contact, 60 h self-study)	Frequency	Annually, summer semester	Duration	1 semester
Module type	Elective	Language	English	Module coordinator	Prof. Dr.-Ing. Kaspar Rehnitz

Formal prerequisites: Successful completion of IAI-M-02

Recommended prior knowledge: None

Learning outcomes

On successful completion of the module, students are able to

- construct a reproducible training and deployment pipeline
- monitor deployed models for data and concept drift
- define rollback and retraining criteria

Module content

- Experiment tracking and model registries
- Continuous integration for model artefacts
- Deployment patterns: shadow, canary, blue-green
- Drift monitoring and alerting
- Governance and model documentation

Teaching and learning methods: Lecture (1 SWS) and computer laboratory (1 SWS)

Form of assessment: PJ — Project work

Requirements for the award of credit points: Submission and acceptance of the project work.

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system; laboratory equipment and experimental set-ups; computer workstations and specialist software

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Huyen: Designing Machine Learning Systems, O'Reilly

Usability: Elective module of the M.Sc. Industrial Artificial Intelligence.

Module description IAI-M-17 — coordinator: Prof. Dr.-Ing. Kaspar Rehnitz, Faculty of Engineering and Computer Science.

IAI-M-18 Robotics Perception

Semester	2	ECTS	3	SWS	2
Workload	90 h (30 h contact, 60 h self-study)	Frequency	Every second year, summer semester	Duration	1 semester
Module type	Elective	Language	English	Module coordinator	Prof. Dr.-Ing. Anselm Bierbaum

Formal prerequisites: Successful completion of IAI-M-07

Recommended prior knowledge: IAI-M-08

Learning outcomes

On successful completion of the module, students are able to

- fuse data from heterogeneous sensors for robot state estimation
- implement pose estimation for grasping tasks
- assess perception reliability in collaborative robot settings

Module content

- Coordinate frames and transformations
- Depth sensing: stereo, structured light, time of flight
- Point cloud registration
- 6-DoF pose estimation
- Sensor fusion and filtering
- Perception requirements for collaborative operation

Teaching and learning methods: Lecture (1 SWS) and robot laboratory (1 SWS)

Form of assessment: M20 — Oral examination, approx. 20 minutes

Requirements for the award of credit points: Passing the oral examination (M20).

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system; laboratory equipment and experimental set-ups

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Thrun, Burgard, Fox: Probabilistic Robotics, MIT Press

Usability: Elective module of the M.Sc. Industrial Artificial Intelligence.

Note: Offered biennially. Next offering summer semester 2026.

Module description IAI-M-18 — coordinator: Prof. Dr.-Ing. Anselm Bierbaum, Faculty of Engineering and Computer Science.

IAI-M-19 German as a Foreign Language A2

Semester	2	ECTS	3	SWS	4
Workload	90 h (60 h contact, 30 h self-study)	Frequency	Annually, summer semester	Duration	1 semester
Module type	Elective	Language	German	Module coordinator	Language Centre, Dr. Ilka Wendehorst

Formal prerequisites: Successful completion of IAI-M-06 or evidence of German at level A1

Recommended prior knowledge: None

Learning outcomes

On successful completion of the module, students are able to

- understand sentences and frequently used expressions related to areas of immediate relevance
- communicate in simple and routine tasks in the workplace
- describe aspects of their background and immediate environment

Module content

- Perfect and preterite tenses
- Dative case; prepositions
- Subordinate clauses
- Workplace and technical everyday vocabulary
- Short reports and applications

Teaching and learning methods: Language course (4 SWS)

Form of assessment: PF — Portfolio comprising at least three components

Requirements for the award of credit points: Portfolio comprising a written test, an oral interaction task and a learning diary.

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Course materials provided by the Language Centre

Usability: Elective module of the M.Sc. Industrial Artificial Intelligence.

***Note:** Taught entirely in German. Unusual workload structure: 4 SWS contact time against 3 ECTS.*

Module description IAI-M-19 — coordinator: Language Centre, Dr. Ilka Wendehorst, Language Centre.

IAI-M-20 Innovation and Technology Management

Semester	2	ECTS	3	SWS	2
Workload	90 h (30 h contact, 60 h self-study)	Frequency	Annually, summer semester	Duration	1 semester
Module type	Elective	Language	English	Module coordinator	Prof. Dr. Marit Sonnleitner

Formal prerequisites: None

Recommended prior knowledge: None

Learning outcomes

On successful completion of the module, students are able to

- assess the maturity of a technology against adoption criteria
- structure an innovation process in a manufacturing enterprise
- evaluate intellectual property strategies for AI-based products

Module content

- Technology readiness assessment
- Innovation processes and stage-gate models
- Business models for data-driven products
- Intellectual property and trade secrets in AI
- Technology roadmapping

Teaching and learning methods: Seminar (2 SWS)

Form of assessment: HA — Term paper

Requirements for the award of credit points: Term paper of 15 to 20 pages.

Media: Board and presentation slides; teaching materials provided electronically via the university learning management system

Admission requirement: No requirements beyond registration for the examination.

Literature:

- Tidd, Bessant: Managing Innovation, Wiley

Usability: Elective module of the M.Sc. Industrial Artificial Intelligence.

Module description IAI-M-20 — coordinator: Prof. Dr. Marit Sonnleitner, Faculty of Engineering and Computer Science.